

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2023-24

Zoology

Paper: ZOB-101 — Animal Diversity and Animal Form and Function

Time: Three hours

Full Marks: 70

Note: Answer three questions from Section-A (including Question No. 1 which is compulsory) and three questions from Section-B (including Question No. 1 which is compulsory). Use a separate answer book for each section. The figures in the right-hand margin indicate marks.

SECTION – A
(Animal Diversity) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Give a brief justification for each: [2 × 2 = 4]
- (i) Cnidoblasts are the stinging cells found in Ctenophora.
 - (ii) *Archaeopteryx* is the connecting link between reptiles and birds.
- (b) Define the following: [3 × 1 = 3]
- (i) Amebocytes
 - (ii) Phasmids
 - (iii) Paedogenesis.
- (c) Differentiate between the following: [2 × 2 = 4]
- (i) Univalvia and Bivalvia
 - (ii) Tentaculata and Nuda.

Question 2. Answer the following:

- (a) Explain why birds are called as glorified reptiles. [6]
- (b) Give a comparative account of adult and larval form of an ascidian (*Herdmania*). [6]

Question 3. Answer the following:

- (a) Classify class Mammalia *or* Amphibia up to their orders giving salient features and examples of each. [6]
- (b) Classify phylum Mollusca *or* Echinodermata up to their classes giving characteristic features and suitable examples of each class. [6]

Question 4. Write short notes on the following:

[4 × 3 = 12]

- (a) Parasitic adaptations in Platyhelminthes
- (b) Neornithes
- (c) Spicules in Porifera
- (d) Retrogressive metamorphosis.

SECTION – B
(Animal Form and Function) (Marks: 35)

Question 1. Answer the following parts:

- (a) State whether the following statements are True or False. Justify your answer in 2–3 sentences each:

[4 × 2 = 8]

- (i) Discontinuous feeders are active and mobile in nature.
- (ii) Freshwater fishes have aglomerular kidney.
- (iii) Complete separation of ventricle facilitates double circulation.
- (iv) The myelin sheath is an extension of a cell organelle.

- (b) Define the following:

[3 × 1 = 3]

- (i) Deposit feeding
- (ii) Sequential hermaphroditism
- (iii) Venous heart.

Question 2. Answer the following:

- (a) With the help of diagrams, explain various methods of asexual reproduction. Comment on its advantages and disadvantages. [6]
- (b) Describe the feeding mechanisms in herbivorous and carnivorous animals citing suitable examples. [6]

Question 3. Answer the following:

- (a) Describe the structure and function of fish gills. With the help of diagram, explain the mechanism of counter current gas exchange. [6]
- (b) Describe the structure and function of protonephridial excretory system with suitable diagrams and examples. [6]

Question 4. Write short notes on the following:

[3 × 4 = 12]

- (a) Otolith
- (b) Cranial nerves
- (c) Cephalization.

[Click Here](#)

[Click Here](#)

[Click Here](#)